



Soft Actor–Critic based active disturbance rejection path following control for unmanned surface vessel under wind and wave disturbances

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ARTICLE INFO

Keywords:

Path following control
Wind and wave disturbances
Linear active disturbance rejection control
Soft Actor–Critic
Reinforcement learning
Parameter optimization

ABSTRACT

The research of unmanned surface vessel (USV) path tracking has always been of great interest in maneuvering control. In the course of navigation, a USV will inevitably be affected by disturbances such as wind and waves, or the USV's operating conditions will suddenly change. To overcome the impact of these disturbances or uncertainties is of great significance to its safe and efficient navigation. In this paper, the three-degree-of-freedom (DOF) manipulative modeling group (MMG) model is used to express the dynamic characteristics of a USV so that its motion process is closer to its real states. We consider the underactuated control problem of using rudder angles to realize the tracking control of the USV's position and heading angle. To this end, we propose a guidance law and design a linear active disturbance rejection controller (LADRC), and the stability of the guidance law and controller are analyzed. This paper combines control and decision-making in reinforcement learning and proposes an LADRC control strategy based on soft actor–critic (SAC) algorithm to realize the adaptive control of USV path tracking. The effectiveness of the proposed method is verified by line and circle under wind and wave environments.

1. Introduction

With the development of ocean engineering, the demand for unmanned surface vehicle (USV) maneuvering movement is gradually increasing. Quick and accurate adjustment of a USV's movement is a prerequisite to ensure its safety and economy. The path tracking control of USVs has recently attracted great interest. During navigation, ships are mainly controlled by human experience, and accidents caused by human errors are not rare. Therefore, increasing attention is paid to the autonomous movement of USV.

Most conventional surface vehicles under study have underactuated characteristics, where the USV's control variables are the rudder and propeller, and the output includes longitudinal position, lateral position, and heading angle. The path tracking system generally includes the three subsystems of guidance, navigation, and control, so that the USV's target point, current position point, and control variable can be obtained (Zhao et al., 2019). Therefore, the difficulty of underactuated USV lies in the design of the guidance law and controller. The three main guidance methods are point-to-point navigation (PPN) (Li et al., 2008), logical virtual ships (LVS) (Zhang et al., 2015; Sun et al., 2017), and line-of-sight (LOS) guidance (Yu et al., 2020; Fossen and Lekkass,

2017). The first two methods use tracking error to design a virtual control law, and the last regards the desired path as a straight line between adjacent track points, thereby converting the tracking of the position to the control of the heading angle. Improved guidance laws based on LOS include error-constrained LOS (ECLOS) (Zheng et al., 2018), surge-guided LOS (SLOS) (Wang et al., 2019), and compound LOS (CLOS) (Miao et al., 2017).

The problems to be overcome in the control system are uncertain or unknown USV parameters, environmental disturbances, and actuator saturation. The most widely used controller methods are dynamic surface control (DSC) (Chwa, 2011; Liu et al., 2017), sliding mode control (SMC) (Hao et al., 2019), adaptive control (Zhang and Zhang, 2015; Do and Pan, 2004; Do et al., 2004), fuzzy control (Nie and Lin, 2019), and backstepping control (Xu et al., 2021; Zheng and Zou, 2016). However, for a USV model with nonlinearity and uncertainty, such a controller will be complicated and difficult to apply. It is desirable to have such a controller that can eliminate the adverse effects of unknown disturbances even when the model information of the controlled object is unknown. Active disturbance rejection control (ADRC), as proposed by Han (2003), has the required characteristics,

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<https://doi.org/10.1016/j.oceaneng.2022.110631>

Received 14 September 2021; Received in revised form 19 December 2021; Accepted 14 January 2022

Available online 5 February 2022

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but is difficult for engineering applications since many parameters must be adjusted. Gao (2013, 2014) developed linear ADRC (LADRC), whose linear extended state observer (LESO) and PD control respectively estimate and eliminate disturbance. LADRC has been applied to control applications such as hydraulic systems (Jiang et al., 2017; Zhuang et al., 2020), unmanned aerial vehicles (UAVs) (Sun et al., 2020; Tao et al., 2016), and power systems (Huang et al., 2021). There has been research of USV path planning controlled by LADRC. Li et al. (2013) combined LADRC with SMC to realize longitudinal path tracking. Tao et al. (2019) applied LADRC to a towing system to simultaneously realize the path tracking control of a tug and drilling platform, but considered only straight-line tracking.

Parameter adjustment is non-negligible in controller design, which will directly affect the control effect. There are fixed- and adaptive-parameter tuning methods. The first type includes heuristic algorithms such as particle swarm optimization (PSO) (Grag, 2016), genetic algorithms (GAs) (Zhou et al., 2018), and sine cosine algorithms (SACs) (Chen et al., 2020), which obtain a set of optimal parameters in a specific situation. The other type uses fuzzy control (Sun et al., 2021) or a neural network (Liu et al., 2021) to obtain the adaptive parameters. However, fuzzy control relies on human experience, and neural networks have no decision-making ability, and can only get a specific mapping relationship, so they may not cope well with sudden environmental changes. There is a need for a parameter tuning algorithm that does not rely on models or human experience, has intelligent ideas, and can adaptively select optimal parameters according to environmental changes.

Reinforcement learning (RL) solves sequential decision-making problems, allowing the environment to obtain action values that are useful for the entire goal process in a certain state by designing an agent. According to the qualitative classification of state and action, RL typically involves Q learning (Yuan et al., 2019) with discrete action and state spaces, a deep Q-network (DQN) (Sewak, 2019) with a discrete action space and continuous state space, or a deep deterministic policy gradient (DDPG) (Qiu et al., 2019) with continuous action and state spaces. The first two methods are suitable for situations with limited actions like mazes (Tijmsma et al., 2016) and chess (Schrittwieser et al., 2020). DDPG has poor convergence due to its extreme brittleness and hyperparameter sensitivity (Fu et al., 2021). Soft actor-critic (SAC) (Haarnoja et al., 2018) overcomes these shortcomings by adding maximum entropy to the reward function to improve exploration ability. RL has achieved certain results in USV path tracking control. Zhao et al. (2020) chose rudder and propeller as the action value and DQN as the controller to realize a USV's path tracking control. Wu et al. (2020) used deep reinforcement learning to achieve autonomous navigation and obstacle avoidance of a USV. These RL-based path tracking control methods are almost completely separated from the control field, which is not conducive to the development of smart controllers.

This paper proposes an intelligent USV path tracking controller based on SAC-optimized LADRC. A USV model based on the manipulative modeling group (MMG) equation is built. A new guidance law is constructed, and an LADRC path following controller is designed, where SAC is applied to obtain its real-time parameters. The stability of the guidance law and controller are given. Straight line and circle tracking simulation experiments are developed to confirm the effectiveness of the control method.

The rest of this paper is organized as follows. Section 2 presents a dynamic model of ships. Section 3 describes the guidance law and design of the LADRC controller and provides a theoretical proof. Section 4 presents the SAC-based parameter optimization algorithm. Section 5 provides simulation results to illustrate the effectiveness of the method. Section 6 relates our conclusions and suggestions for future work.

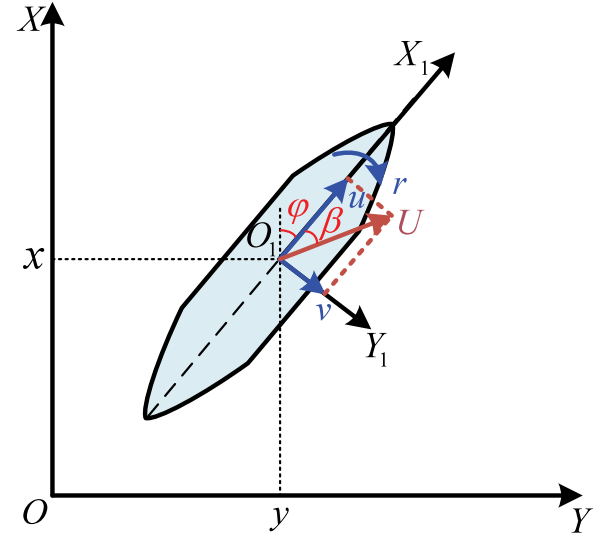


Fig. 1. Schematic diagram of USV coordinates.

2. USV dynamic motion based on MMG

USV motion is a complex process whose precise mathematical modeling is challenging; a straightforward model cannot describe the motion characteristics of ships. This paper ignores heave, pitch, and roll motions. Only the slew, sway, and yaw motions are considered. A three degree-of-freedom (DOF) USV MMG model is established, where the hydrodynamic force and torque acting on the hull, propeller, and rudder are expressed as follows.

2.1. Plane motion equation of USV

The movement of a USV is usually expressed in inertial and fixed coordinate systems OXY and $O_1X_1Y_1$, respectively, as shown in Fig. 1, where O_1 is at the center of gravity of the USV; (x, y) are the coordinates of the USV's position; u , v , and r are the forward speed, lateral movement speed, and yaw angular velocity, respectively; the angle between the X_1 and X axes is specified as the heading angle φ ; and the angle between the combined velocity U of the USV and the X_1 axis is defined as the drift angle β , shown as

$$\beta = \arctan(v/u). \quad (1)$$

Assuming that the USV is a rigid body, its position and heading angle can be derived as

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\varphi} \end{pmatrix} = \begin{pmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \\ r \end{pmatrix} \quad (2)$$

The position and heading angle of the USV can be obtained from u , v , and r . The three-DOF MMG model (Li et al., 2013) is used to solve for these variables, and is expressed as

$$\begin{cases} (m + m_x) \dot{u} - (m + m_y) vr = X_H + X_P + X_R + X_E \\ (m + m_y) \dot{u} + (m + m_x) ur = Y_H + Y_P + Y_R + Y_E \\ (I_{zz} + J_{zz}) \dot{r} = N_H + N_P + N_R + N_E \end{cases} \quad (3)$$

where m and I_{zz} are the USV's mass and moment of inertia, respectively. m_x, m_y denote the additional inertial mass on the x and y axes, respectively. And J_{zz} represents the additional moment of inertia of the USV in deep water. The right side of the equation has the forces X and Y acting on the X_1 and Y_1 axes, respectively, and the moment N , where subscripts H , P , R , and E represent the hull, propeller, rudder, and external environment, respectively.

2.2. Calculation of forces and moments

In this section, Eq. (3) can be solved when the additional mass, additional moment of inertia, and force and moment of force are obtained.

2.2.1. Hull viscous fluid force and moment

The hydrodynamic force and moment of the hull are calculated as

$$\begin{cases} X_H = X(u) + X_{vv}v^2 + X_{vr}vr + X_{rr}r^2 + X_{vvv}v^3 \\ Y_H = Y_vv + Y_r r + Y_{|v|v}|v|v + Y_{|r|r}|r|r + Y_{vvr}v^2r + Y_{vrr}vr^2 \\ N_H = N_vv + N_r r + N_{|v|v}|v|v + N_{|r|r}|r|r + N_{vvv}v^3 + N_{vrr}vr^2 \end{cases} \quad (4)$$

where $X(u)$ is the hull resistance of the USV during direct voyage, and X_{vv} , X_{vr} , X_{rr} , X_{vvv} , Y_v , Y_r , $Y_{|v|v}$, $Y_{|r|r}$, Y_{vvr} , Y_{vrr} , N_v , N_r , $N_{|v|v}$, $N_{|r|r}$, N_{vvv} , N_{vrr} are maneuverable hydrodynamic derivatives, where subscripts v and r indicate position and rotation, respectively. Detailed formulas are omitted due to space limitations and can be found in the literature (Jia and Yang, 1999).

2.2.2. Propeller force and moment

The lateral force and moment produced by the propeller are very small at constant speeds, so we only calculate the longitudinal thrust,

$$\begin{cases} X_P = (1 - t_p) \rho n^2 D_p^4 K_T (J_P) \\ Y_P = 0 \\ N_P = 0 \end{cases} \quad (5)$$

where D_p , n , and J_P are the diameter, speed, and advanced speed coefficient, respectively, of the propeller; t_p is the number of propeller thrust deratings; ρ is the sea water density; and K_T is the propeller thrust coefficient.

2.2.3. Rudder force and moment

The rudder force is expressed as

$$\begin{cases} X_R = -(1 - t_R) F_N \sin \delta \\ Y_R = -(1 + a_H) F_N \cos \delta \\ N_R = -(x_R + a_H x_H) F_N \cos \delta \end{cases} \quad (6)$$

where F_N and δ are the positive pressure and angle, respectively, of the rudder; t_R is the derating of rudder resistance; a_H is the coefficient of the influence of steering on the lateral force of the hull; and x_R and x_H are the longitudinal coordinate of the rudder center and distance from the center of the lateral force of the hull to the center of gravity of the USV, respectively.

2.3. Disturbance dynamic model

In actual navigation, a USV may encounter unexpected situations such as wind or waves, which may cause deviation from the established path. Therefore, we consider their influence on the USV during its movement. The forces and moments acting on the USV are calculated as follows.

2.3.1. Wind

The relative speed of USV motion when affected by a steady wind with direction α_w and speed U_w is

$$\begin{cases} u_{rw} = u + U_w \cos(\alpha_w - \varphi) \\ v_{rw} = v + U_w \sin(\alpha_w - \varphi) \end{cases} \quad (7)$$

and the wind pressure and moment acting on the USV are

$$\begin{cases} X_w = -\frac{1}{2} \rho_a A_f (u_{rw}^2 + v_{rw}^2) C_{wx}(\alpha_w) \\ Y_w = \frac{1}{2} \rho_a A_s (u_{rw}^2 + v_{rw}^2) C_{wy}(\alpha_w) \\ N_w = \frac{1}{2} \rho_a A_s L (u_{rw}^2 + v_{rw}^2) C_{wn}(\alpha_w) \end{cases} \quad (8)$$

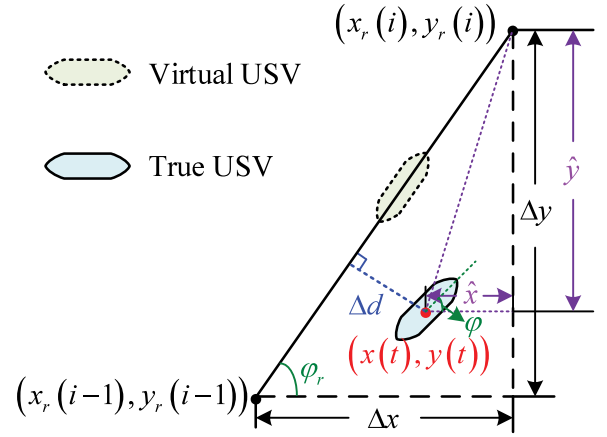


Fig. 2. Schematic diagram of USV path tracking.

where ρ_a is the air density; A_f and A_s are the orthographic projection and measured projection areas, respectively, above the USV's waterline; L is the length of the USV; C_{wx} and C_{wy} are the wind coefficients on the x- and y-axes, respectively; and C_{wn} is the wind moment coefficient around the z-axis.

2.3.2. Wave

The relative speed of motion of a USV affected by a steady wave with velocity U_{wa} and direction α_{wa} can be obtained as

$$\begin{cases} u_{rwa} = u + U_{wa} \cos(\alpha_{wa} - \varphi) \\ v_{rwa} = v + U_{wa} \sin(\alpha_{wa} - \varphi) \end{cases} \quad (9)$$

and the dynamic equation of a USV affected by water flow is

$$\begin{cases} \dot{x} = u \cos \varphi - v \sin \varphi + U_{wa} \cos \alpha_{wa} \\ \dot{y} = u \sin \varphi + v \cos \varphi + U_{wa} \sin \alpha_{wa} \\ \dot{\varphi} = r \\ (m + m_x) \dot{u} - (m + m_y) vr = X_H + X_P + X_R + (m + m_x) r (v_{rwa} - v) \\ (m + m_y) \dot{u} + (m + m_x) ur = Y_H + Y_P + Y_R - (m + m_y) r (u_{rwa} - u) \\ (I_{zz} + J_{zz}) \dot{r} = N_H + N_P + N_R + 0 \end{cases} \quad (10)$$

3. Design of path following controller

It can be seen from the above model that the USV motion includes linear motion on the x- and y-axes and rotational motion around the z-axis. The USV's speed is driven by the speed of the propeller n , and the heading angle φ is changed by the rudder angle δ . We consider the USV's path tracking control under a constant propeller speed, i.e., the USV will move on a predetermined path by controlling the rudder angle.

3.1. Guidance law

The USV moves with the reference waypoint during the voyage. Taking a short straight line as an example, as shown in Fig. 2, assuming that the reference waypoint is generated by a virtual USV, $(x_r(i-1), y_r(i-1))$ and $(x_r(i), y_r(i))$ are its positions at this time and the previous time, respectively, and $(x(t), y(t))$ is the current real USV position.

According to Fig. 2,

$$\begin{cases} \Delta x = x_r(i) - x_r(i-1) \\ \Delta y = y_r(i) - y_r(i-1) \\ \hat{x} = x_r(i) - x(t) \\ \hat{y} = y_r(i) - y(t) \end{cases} \quad (11)$$

and we can derive that

$$\begin{cases} \Delta d = \frac{\hat{x}\Delta y - \hat{y}\Delta x}{\sqrt{\Delta x^2 + \Delta y^2}} \\ \varphi_r = \arctan(\Delta y/\Delta x) \end{cases} \quad (12)$$

where Δd is the tracking error between the USV and the straight path traveled by the virtual USV, and φ_r is the heading course of the virtual USV. When $\Delta d \approx 0$ and $(\varphi - \varphi_r) \approx 0$, the true USV will track the path of the virtual USV. In other words, we need to use the rudder angle to control the two kinds of variables. For this underactuated control system, we design a nonlinear combination as

$$y_o = c_1 \tanh(c_0 \Delta d) + (\beta + \varphi - \varphi_r). \quad (13)$$

where the constant parameters c_0 and c_1 are positive. The drift angle β is used to compensate for lateral drift under disturbance, and the hyperbolic tangent function is used to constrain the range of track deviation. Then in steady state, $y_o = 0$.

3.2. Design of LADRC

It is expected that the USV can follow the planned path by controlling the rudder angle. Here, in Eq. (13), $y_o = 0$ is the control target of path tracking, which is equivalent to $\varphi = \varphi_r - c_1 \tanh(c_0 \Delta d)$. Then φ can be regarded as the virtual controlled variable that stabilizes Δd . In other words, the proposed guidance law transforms the path tracking problem into the control of the USV's heading angle. We use LADRC to obtain the rudder angle to realize USV path tracking. The design of the controller does not need the precise information of the controlled plant, but only needs to determine the order of the controlled system.

According to Eq. (3), we can derive from the heading angle φ and rudder angle δ that

$$\begin{aligned} \ddot{\varphi} &= \frac{1}{I_{zz} + J_{zz}} (N_H + N_P + N_R(\delta) + N_E) \\ &= \frac{1}{I_{zz} + J_{zz}} (N_H + N_P + N_E) + \frac{1}{I_{zz} + J_{zz}} (- (x_R + a_H x_H) F_N \cos \delta). \end{aligned} \quad (14)$$

According to Eq. (13), $\ddot{y}$ is expressed as

$$\begin{aligned} \ddot{y}_o &= \ddot{\varphi} + c_0 c_1 \Delta \ddot{d} (1 + \tanh^2(c_0 \Delta d)) \\ &\quad + 2c_0^2 c_1 \Delta \dot{d}^2 \tanh(c_0 \Delta d) (1 - \tanh^2(c_0 \Delta d)) \end{aligned} \quad (15)$$

where $\Delta \ddot{d}$ can be derived according to

$$\Delta \ddot{d} = \frac{u + v}{\sqrt{u^2 + v^2}} \sin(\varphi - \varphi_r) + \dot{\varphi} \sqrt{u^2 + v^2} \cos(\varphi - \varphi_r) \quad (16)$$

Combining Eq. (14)–(16), Eq. (13) can be organized into a second-order system,

$$\ddot{y}_o = f(u, v, \varphi, \dot{\varphi}, \tau(\delta)) + b_0 \delta \quad (17)$$

where $f(u, v, \varphi, \dot{\varphi}, \tau(\delta))$ is the total disturbance, including inside and outside disturbance like wind or waves, $\tau(\delta)$ is the functional representation of δ , and b_0 is a constant. The core idea of LADRC is to estimate the value of $f(u, v, \varphi, \dot{\varphi}, \tau(\delta))$ and eliminate it. Therefore, the following combination of a linear expanded state observer (LESO) and PD control law is designed.

Define the system states as $x_1 = y_o$, $x_2 = \dot{y}_o$, $x_3 = f(u, v, \varphi, \dot{\varphi}, \tau(\delta))$. Then the LESO can be designed as

$$\begin{cases} \dot{z}_1 = z_2 + \beta_1 (x_1 - z_1) \\ \dot{z}_2 = z_3 + \beta_2 (x_1 - z_1) + b_0 \delta \\ \dot{z}_3 = \beta_3 (x_1 - z_1) \end{cases} \quad (18)$$

where z_1 , z_2 , and z_3 are estimates of states x_1 , x_2 , and x_3 , respectively; and β_1 , β_2 , β_3 are observer gains, which will affect the accuracy of state estimation. The observer pole $-\omega_o$ is used to adjust the observer gains: $\beta_1 = 3\omega_o$, $\beta_2 = 3\omega_o^2$, $\beta_3 = \omega_o^3$ (Gao, 2014). Therefore, an appropriate ω_o can guarantee that $z_3 \approx f(u, v, \varphi, \dot{\varphi}, \tau(\delta))$.

To eliminate the influence of disturbance on the system, the control law is designed as

$$\delta = \frac{k_p (\hat{y}_o - z_1) + k_d (\dot{\hat{y}}_o - z_2) + \ddot{\hat{y}}_o - z_3}{b_0} \quad (19)$$

where $\hat{y}_o$ is the desired value of y_o , and k_p, k_d are the feedback control gains. Similarly, the adjustment of k_p, k_d is converted to the adjustment of the state feedback control pole $-\omega_c$: $k_p = \omega_c^2$, $k_d = 2\omega_c$.

3.3. Stability analysis

We analyze the stability of the guidance law and controller.

Theorem 1. The convergence of the guidance law shown in Eq. (13) can be guaranteed by adjusting the parameter c_1 .

Proof of Theorem 1. We construct the Lyapunov function as $V = \frac{1}{2} \Delta d^2$. After derivation, we can get

$$\dot{V} = \Delta d \Delta \dot{d} \quad (20)$$

Substituting Eq. (11) in Eq. (12), we have

$$\Delta d = (x_r(i) - x(t)) \sin \varphi_r - (y_r(i) - y(t)) \cos \varphi_r \quad (21)$$

Then, from Eqs. (1), (2), and (21), we have

$$\begin{aligned} \Delta \dot{d} &= -\dot{x}(t) \sin \varphi_r + \dot{y}(t) \cos \varphi_r \\ &= -(u \cos \varphi - v \sin \varphi) \sin \varphi_r + (u \sin \varphi + v \cos \varphi) \\ &= \sqrt{u^2 + v^2} \sin(\beta + \varphi - \varphi_r) \end{aligned} \quad (22)$$

Substituting Eq. (13) into Eq. (22) can get the following expression

$$\Delta \dot{d} = \sqrt{u^2 + v^2} \sin(y_o - c_1 \tanh(c_0 \Delta d)) \quad (23)$$

Eq. (20) is derived as

$$\dot{V} = \Delta d \sqrt{u^2 + v^2} \sin(y_o - c_1 \tanh(c_0 \Delta d)) \quad (24)$$

Then we can rearrange Eq. (24) as

$$\dot{V} = \Delta d \sqrt{u^2 + v^2} \sin(y_o - c_1 \tanh(c_0 \Delta d)) \quad (25)$$

Let $y_o = 0$. Then $\dot{V} = \Delta d \sqrt{u^2 + v^2} \sin(-c_1 \tanh(c_0 \Delta d))$. There are three situations.

Case 1. $\Delta d = 0$. Obviously, we have $\dot{V} = 0$.

Case 2. $\Delta d > 0$. If $c_0 > 0$ and $c_1 > 0$, then $-c_1 < -c_1 \tanh(c_0 \Delta d) < 0$, and there is $0 < c_1 < \pi$, so that $\dot{V} < 0$.

Case 3. $\Delta d < 0$. Similarly, when $0 < c_1 < \pi$, $\dot{V} < 0$.

Assumption 1. Let $f(u, v, \varphi, \dot{\varphi}, \tau(\delta)) = f + b_0 \varsigma_b \delta$, where ς_b is the perturbation of b_0 . f is differentiable and

$$|\dot{f} + b_0 \varsigma_b \dot{\delta}| \leq h_f \quad (26)$$

Theorem 2. The estimation error e_1, e_2 and tracking error μ_1, μ_2, μ_3 are bounded and, under the premise of $1 - 2\lambda_{\max}(P) |\varsigma_b| \|T\| > 0$,

$$\lim_{t \rightarrow \infty} \|E\| \leq \frac{2\lambda_{\max}^2(P) h_f}{\lambda_{\min}(P) [1 - 2\lambda_{\max}(P) |\varsigma_b| \|T\|]}$$

where $E = [e_1, e_2, \mu_1, \mu_2, \mu_3]^T$. $\lambda_{\max}(P)$ and $\lambda_{\min}(P)$ are the maximum and minimum eigenvalues, respectively, of P .

Proof of Theorem 2. Define $e_i = y_r^{(i-1)} - x_i$, ($i = 1, 2$), $\mu_j = x_j - z_j$, ($j = 1, 2, 3$). Then, the state space equation can be organized as

$$\dot{E} = A_e E + H \quad (27)$$

$$\text{where } A_e = \begin{bmatrix} A & B \\ 0 & D \end{bmatrix}, H = [0_{4 \times 1} \quad \varsigma_b T E + \dot{f} + \varsigma_b \ddot{y}_r]^T,$$

$$A = \begin{bmatrix} 0 & 1 \\ -\omega_c^2 & -2\omega_c \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ -\omega_c^2 & -2\omega_c \end{bmatrix}, D = \begin{bmatrix} -3\omega_o & 1 & 0 \\ -3\omega_o^2 & 0 & 1 \\ -\omega_o^3 & 0 & 0 \end{bmatrix},$$

$$T = [-2\omega_c^3, -3\omega_c^2, -3\omega_o\omega_c^2 - 6\omega_c\omega_o^2 - 2\omega_c^2 - \omega_o^3, -3\omega_c^2, 0].$$

It can be seen that A_e is a Hurwitz matrix and there exists a positive definite matrix P such that $PA_e + A_e^T P = -I$. Then there is a Lyapunov function,

$$V = E^T P E \quad (28)$$

whose derivative can be derived as

$$\begin{aligned} \dot{V} &= \frac{\partial V}{\partial E} A_e E + \frac{\partial V}{\partial \mu_3} (\zeta_b T E + \dot{f} + \zeta_b \ddot{r}) \\ &\leq -\|E\|^2 + \frac{\partial V}{\partial E} (|\zeta_b| \|T\| \|E\| + \dot{f} + \zeta_b \ddot{r}) \\ &\leq -\|E\|^2 + 2\lambda_{\max}(P) \|E\| (|\zeta_b| \|T\| \|E\| + h_f) \end{aligned} \quad (29)$$

For Eq. (28), there is $\lambda_{\min}(P) \|E\|^2 \leq V \leq \lambda_{\max}(P) \|E\|^2$. Thus, Eq. (29) can be further sorted to obtain the inequality

$$\dot{V} \leq -\frac{V}{\lambda_{\max}(P)} (1 - 2\lambda_{\max}(P) |\zeta_b| \|T\|) + 2\lambda_{\max}(P) \|E\| h_f \quad (30)$$

Let $\sqrt{V} = W$. Then $\dot{W} = \frac{\dot{V}}{2\sqrt{V}}$, and Eq. (30) is equivalent to

$$\dot{W} \leq -\frac{W}{2\lambda_{\max}(P)} (1 - 2\lambda_{\max}(P) |\zeta_b| \|T\|) + \frac{\lambda_{\max}(P) h_f}{\sqrt{\lambda_{\min}(P)}} \quad (31)$$

Then we can use the Gronwall–Bellman inequality to obtain that

$$\begin{aligned} W &\leq -\left(W(t_0) - \frac{2\lambda_{\max}^2(P) h_f}{\sqrt{\lambda_{\min}(P)} (1 - 2\lambda_{\max}(P) |\zeta_b| \|T\|)} \right) \\ &\quad \times e^{-\frac{1-2\lambda_{\max}(P) |\zeta_b| \|T\|}{2\lambda_{\max}(P)} (t-t_0)} \\ &\quad + \frac{2\lambda_{\max}^2(P) h_f}{\sqrt{\lambda_{\min}(P)} (1 - 2\lambda_{\max}(P) |\zeta_b| \|T\|)} \end{aligned} \quad (32)$$

So,

$$\lim_{t \rightarrow \infty} W \leq \frac{2\lambda_{\max}^2(P) h_f}{\sqrt{\lambda_{\min}(P)} (1 - 2\lambda_{\max}(P) |\zeta_b| \|T\|)} \quad (33)$$

Since $W \geq \sqrt{\lambda_{\min}(P)} \|E\|$, we can obtain that

$$\lim_{t \rightarrow \infty} \|E\| \leq \frac{2\lambda_{\max}^2(P) h_f}{\lambda_{\min}(P) (1 - 2\lambda_{\max}(P) |\zeta_b| \|T\|)} \quad (34)$$

Therefore, under the premise that $(1 - 2\lambda_{\max}(P) |\zeta_b| \|T\|) > 0$, the estimation errors and tracking errors are bounded.

4. Design of adaptive controller parameters based on SAC

The SAC algorithm has the decision-making ability of RL and the computational advantage of a neural network, and can be used to solve a system with a continuous state and action. Parameters requiring tuning are ω_o, ω_c, b_0 in the LADRC controller and c_0, c_1 in the guidance law. A reward function can be designed for a system containing the controller as the environment. Through continuous interaction between the environment and the agent, one can find the optimal strategy, i.e., the adaptive parameters.

4.1. Main idea of SAC

RL essentially uses cumulative rewards R_c to guide a complex rule, so the optimal action can be found through the state of the system, where $R_c = \sum_{k=0}^{\infty} \gamma^k r_{k+1}$, and γ reflects the importance of a reward value for a future moment. The establishment of rules is based on the state–action value function,

$$Q_{\pi}(s, a) = E_{\pi} [R_c | S_t = s, A_t = a] \quad (35)$$

which is equivalent to a probabilistic representation of the action corresponding to state s . Obviously, a larger $Q_{\pi}(s, a)$ implies a greater probability of choosing action a .

The neural network approximation method can solve high-dimensional problems that a table cannot solve. Therefore, neural networks are usually used with continuous state and action systems. SAC combines the advantages of value- and policy-based methods in RL, where the former uses a neural network to fit the Q value, $f(s, a; \theta) \approx Q(s, a)$, and the latter uses a neural network to output the distribution of actions in a certain state. Fig. 3 shows the general structure of the above methods, where $s = [s_1, s_2, \dots, s_n]$ and $a = [a_1, a_2, \dots, a_m]$ are the state and action vectors, respectively, of the system.

In the SAC algorithm, a critic network is used to fit $Q_{\pi}(s, a)$, and an actor network can obtain the action distribution closer to the true distribution under the guidance of Q_s , which is expressed as

$$Q_{\pi}(s, a) = \sum_i E_{\pi} [r(s, a) + \alpha H(\pi(\cdot | s)) | S_t = s, A_t = a] \quad (36)$$

where $a \sim \pi(s)$, and $H(\pi(\cdot | s))$ represents the entropy under policy π , which describes the uncertainty of random variables. α can be seen as the weight of entropy.

The ultimate purpose of SAC is to find the optimal policy π^* , i.e., $\pi^* = \arg \max_{\pi} Q_{\pi}(s, a)$. This shows that SAC not only requires the largest cumulative reward, but also tends to find the largest entropy (random variable with equal probability), which can prevent the system from repeatedly selecting the same action and falling into suboptimality. The update expression of Q_s is expressed as

$$Q_s(s_t, a_t) = r(s_t, a_t) + \gamma [Q_s(s_{t+1}, a_{t+1}) - \alpha \log(\pi(\cdot | s_{t+1}))] \quad (37)$$

4.2. Design of SAC

Fig. 4 is a schematic diagram to more clearly show the structure of the SAC algorithm, which consists mainly of the replay buffer, critic network, and actor network. SAC is data-driven and does not rely on model information, so, in the early stage of training, the environment will be used to obtain the data (s_t, a_t, r_t, s_{t+1}) , which are needed to train the agent and will be stored in the replay buffer.

It can be observed from Fig. 4 that there are four critic networks with the same structure, consisting of Q -networks $Q1, Q2$, and two target Q -networks $QTarget1, QTarget2$, whose weights are θ_1, θ_2 and θ'_1, θ'_2 , respectively. It is worth noting that weights θ'_1 and θ'_2 are not updated through training, but are obtained through the exponentially moving average method on the basis of θ_1 and θ_2 ,

$$\theta'_i = \tau \theta_i + (1 - \tau) \theta'_i, (i = 1, 2) \quad (38)$$

where τ is a smoothing factor. The target network is meant to make the network training more stable, drawing on the idea of DQN.

The Q -network weight is updated based on the loss function

$$J(\theta_i) = \frac{1}{2} \left[\min_Q Q_{\pi}(s_t, a_t; \theta_i) - Q_s(s_t, a_t) \right]^2 \quad (39)$$

where minimization is meant to alleviate the overestimation problem, and can be used with the gradient descent method to optimize θ_1 and θ_2 .

The actor network, assuming a weight of ϕ , outputs the action value through the state value. The output of the policy network mentioned above is the distribution $\pi(\cdot | s)$. The action distribution of SAC is no longer concentrated on a certain action, but should be proportional to the exponential distribution of Q_s , i.e., $\pi(a_t | s_t) \propto \exp\{Q_s(s_t, a_t)\}$, which can be derived from Eq. (37). Therefore, the loss function of the actor network is

$$J_{\pi}(\phi) = D_{KL} \left(\pi(\cdot | s_t) \left\| \frac{\exp\{Q_s(s_t, \cdot)\}}{Z(s_t)} \right\| \right) \quad (40)$$

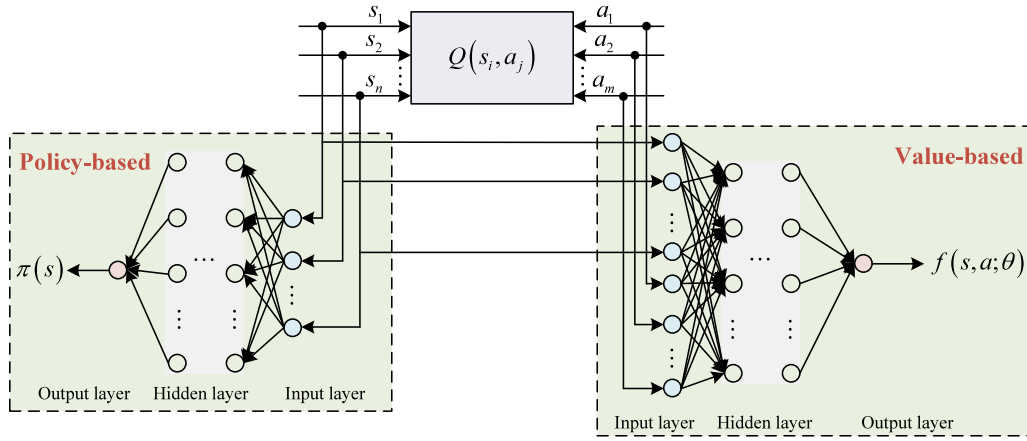


Fig. 3. Structure comparison chart of policy- and value-based methods.

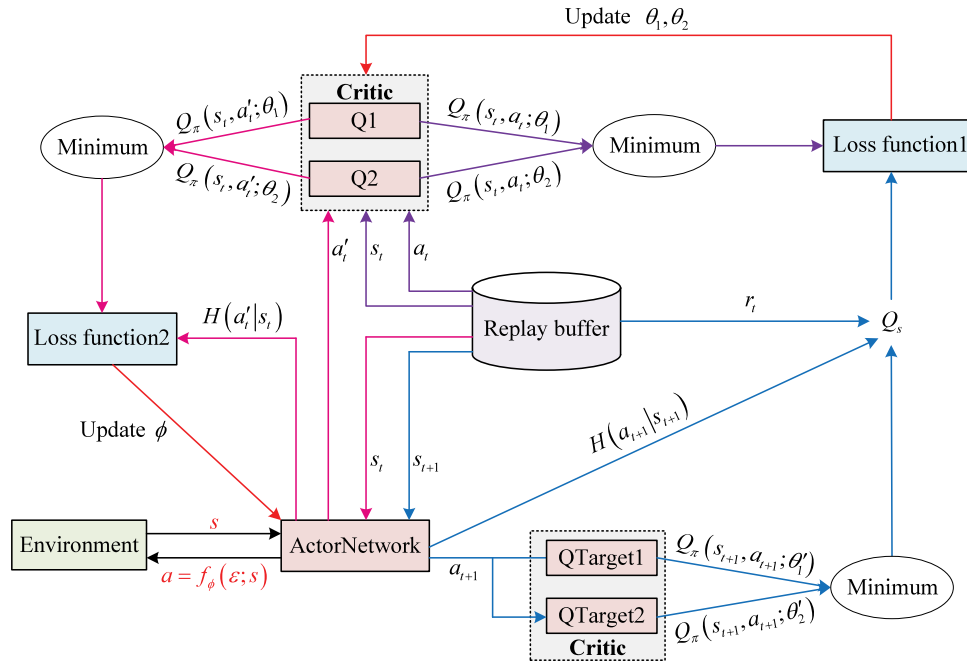


Fig. 4. Structure of SAC.

where D_{KL} is the Kullback–Leibler divergence and $Z(s_t)$ is the log partition function independent of parameter ϕ . Ignoring the constant $Z(s_t)$, which does not affect the gradient, and sorting out Eq. (40), loss function2 in Fig. 4 can be obtained as

$$J_{\pi}(\phi) = \log(\pi_{\phi}(a_t | s_t)) - Q_s(s_t, a_t) \quad (41)$$

where a_t is obtained on the basis of the mean $f_{\phi}^{\mu}(s_t)$ and covariance $f_{\phi}^{\sigma}(s_t)$ of the network output,

$$a_t = f_{\phi}(\epsilon_t; s_t) = f_{\phi}^{\mu}(s_t) + \epsilon_t f_{\phi}^{\sigma}(s_t) \quad (42)$$

where ϵ_t is sampled from a Gaussian distribution.

4.3. Handling of states and actions

In the course of a voyage, a USV may encounter wind or waves, causing it to deviate from the settling path. The parameters of the LADRC controller will directly affect the USV tracking effect. Therefore, it is important to choose appropriate controller parameters to minimize USV tracking error. SAC is mainly used to solve the problem of LADRC parameter adjustment.

To make the agent more robust, the error and its derivative at time k are used to define the state,

$$\begin{cases} s_1(k) = y_o(k) \\ s_2(k) = s_1(k) - s_1(k-1) \end{cases} \quad (43)$$

This shows that at every moment, the system can generate a state vector $s(k) = [s_1(k), s_2(k)]^T$, which can be input to the SAC system.

For the entire controlled system, according to the above description, it can be known that the parameters requiring adjustment are c_0 and c_1 of the guidance law and ω_o , ω_c , and b_0 of the controller. Theorem 1 shows that there exists $0 < c_1 < \pi$, which makes the proposed guidance law converge, and c_0 forces the tracking error Δd and heading angle to have the same magnitude, which has little effect on the convergence of the guidance law, so the two parameters can be manually adjusted. Furthermore, for the controller parameters, the literature (Zhou and Tan, 2019) points out through analysis that b_0 is not an essential factor in suppressing interference, and feedback gain K and observer gain L can compensate for its influence. Therefore, we use SAC to adjust ω_o and ω_c , i.e., the action at time k is selected as $a(k) = [\omega_o(k), \omega_c(k)]^T$.

The definition of states and actions is the first step for the agent to interact with the environment. To enable it to make decisions

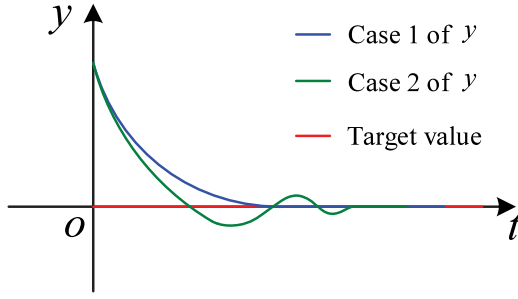
Fig. 5. Schematic diagram of change process of y .

Table 1
Main dimensions of USV model.

Parameter	Value	Parameter	Value
Displacement	14 635 t	Propeller diameter	4.6 m
Length	126 m	Propeller pitch	3.66 m
Width	20.8 m	Rudder area	18.8 m ²
Square factor	0.681	Rudder height	5.68 m
Draught	8 m	Aspect ratio	1.72

according to the desired effect, a reward function is also necessary, and is expected to encourage the USV to sail along the desired path. USV tracking to the desired path is equivalent to $y_o = 0$ in Eq. (13). As shown in Fig. 5, we hope that y can smoothly stabilize to 0, as in case 1, and we try to avoid oscillations in case 2. Observing the changing trend of the curve in Fig. 5, we can find that $y\dot{y} \leq 0$ is what we want. Therefore, the reward function is designed as

$$r(k) = -\tanh(|s_1(k)|) - \text{sign}(s_1(k)s_2(k)) \quad (44)$$

As shown in Fig. 6, the process is divided into data acquisition, network training, and online stages. The first stage does not involve network updates, and only uses the interaction between the environment and the actor network to get the required data. Samples are randomly selected in the second stage, and the network parameters θ and ϕ are updated by Eqs. (39) and (41). Finally, the state value at each moment can be input to the trained network to obtain the optimal action value.

5. Simulation examples

Simulation experiments of USV path following control are carried out, with the USV model parameters shown in Table 1. We use the above method to realize the path tracking control of the model.

According to Theorem 2, a positive definite matrix can be constructed as

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} & p_{15} \\ p_{12} & p_{22} & p_{23} & p_{24} & p_{25} \\ p_{13} & p_{23} & p_{33} & p_{34} & p_{35} \\ p_{14} & p_{24} & p_{34} & p_{44} & p_{45} \\ p_{15} & p_{25} & p_{35} & p_{45} & p_{55} \end{bmatrix} \quad (45)$$

which satisfies $PA_e + A_e^T P = -I$. One can refer to the Appendix for matrix element expression.

In this way, the inequality about ζ_b can be obtained as $|\zeta_b| < \frac{1}{2\lambda_{\max}(P)\|T\|}$, which is a sufficient condition for controller stability. Then, the visualization diagram of ω_o , ω_c , and ζ_b can be established, as shown in Fig. 7.

Fig. 7 gives a rough guide on the choice of parameters. It can be seen that when ω_o and ω_c take a smaller interval, $|\zeta_b|_{\max}$ is relatively large. We set b_0 to 0.00002, and c_0 and c_1 are 0.008 and $3\pi/4$, respectively.

The parameters of the agent are shown in Table 2, and the network structures of the critic and actor are shown in Fig. 8. We use simulations of straight and circular path tracking in wind and wave environments

Table 2
Parameters of SAC.

Parameters	Value
Random mini-batch	$m = 32$
Replay buffer size	$M = 10^6$
Learning rate of critic	$\alpha_1 = 10^{-4}$
Learning rate of actor	$\alpha_2 = 10^{-4}$
Discount factor	$\gamma = 0.9$
Target smooth factor	$\tau = 0.001$
Sample time	0.25 s

to verify the effectiveness of the proposed method, where the initial USV speed is $u = 4$ m/s, $v = 0$ m/s, and $r = 0$ m/s. The propeller speed is a constant 103.4 revolutions per minute.

5.1. Straight path tracking

To verify the effectiveness of the proposed method, the reference path is designed as two line segments from point (0,0) to (3000,3000), and from (3000,3000) to (0,6000). Suppose the initial position and heading angle of the USV are (0,50 m) and 45° , respectively. To demonstrate the anti-disturbance performance of the proposed method, wave and wind disturbances are added at 450 s and 1500 s, respectively. Fig. 9 shows the comparison results between SAC-LADRC and LADRC with fixed parameters of $\omega_o = 2.5$, $\omega_c = 0.025$ on path tracking and the change process of the rudder angle, heading angle, and controller parameters during tracking.

It can be observed that the path obtained by SAC-LADRC and LADRC can both finally stabilize to the planned path. However, the results obtained by SAC-LADRC has a smaller overshoot and a faster speed at the turning area of the path and the area facing with wind and wave disturbance. Then, by observing the rudder and heading angles, it can be seen that when the USV is disturbed by wind and waves, the rudder angle will change to deal with them. It can also be seen from the partial enlargement that, compared to the method proposed in this paper, the LADRC controller with fixed parameters oscillates more obviously when faced with wind and wave disturbances. In addition, in the steady state, ψ and ψ_d are not consistent, which is caused by the existence of drift angle β . In Fig. 9(d), y_o is 0 in the steady state, which numerically demonstrates the effectiveness of the proposed control law. Figs. 9(e) and 9(f) respectively show the changing process of ω_o and ω_c . It should be pointed out that, unlike the Q learning and DQN algorithms in reinforcement learning, the SAC algorithm obtains a distribution of actions, so actions corresponding to the same state may be different. In addition, the parameter change appears to be dramatic because the simulation interval is only 0.25 s.

5.2. Circular path tracking

To show the tracking effect of the proposed method on the curve, we use a circle with center (0 m,0 m) and radius 1000 m as the set path.

5.2.1. Circular path tracking with wind disturbance

Assume that the USV is disturbed by wind with a speed of 5 m/s and direction of 60° starting from 600 s. The initial position and heading angle of the USV are respectively (0 m,1050 m) and 0° , and it moves in a clockwise direction. Figs. 10(a)–10(d) shows the USV path tracking control effect obtained under the action of the controller parameters shown in Figs. 10(e) and 10(f), optimized by SAC. The control effect of the LADRC controller with a fixed parameter of $\omega_o = 0.8$, $\omega_c = 0.008$ is also shown in Figs. 10(a)–10(d).

It can be seen that the rudder angle control variable obtained by SAC-LADRC can well stabilize the USV on the planned path. When the USV is disturbed by wind, the rudder angle makes a clear response and stabilizes the USV on the planned circular path. In addition, the results

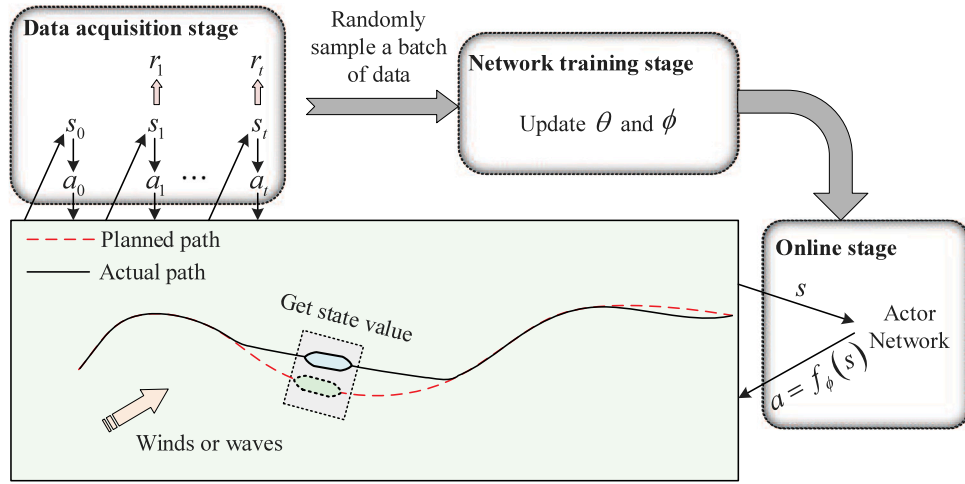


Fig. 6. Schematic diagram of process of USV path tracking LADRC controller based on SAC optimization.

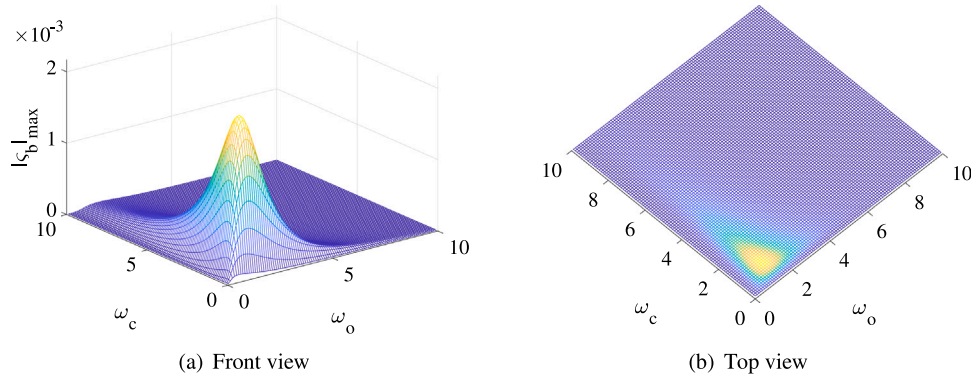


Fig. 7. Relationship diagram of ω_o , ω_c , and $|zeta_b|_{\max}$.

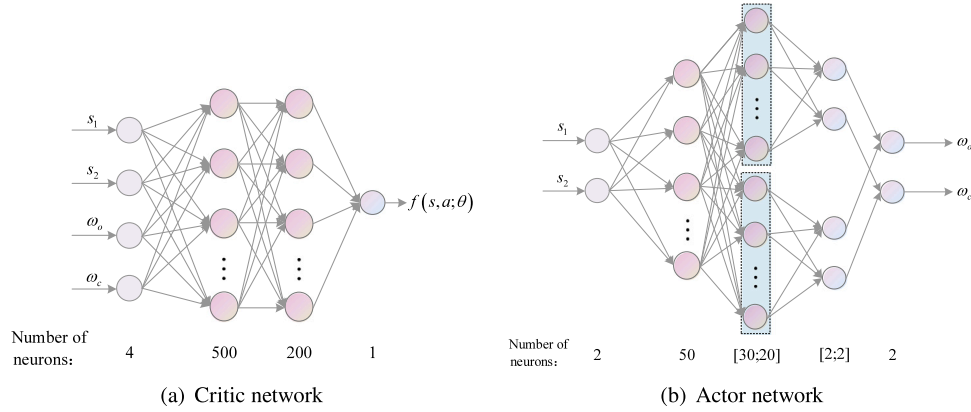


Fig. 8. Structure of critic and actor network.

of Figs. 10(a) and 10(d) reflect to a certain extent that when tracking a curved path, the tracking speed and tracking error obtained by the proposed SAC-LADRC are significantly better than that of the LADRC controller.

5.2.2. Circular path tracking with wind current disturbance

In the course of sailing, waves will also affect the path of the USV. It is assumed that the USV is disturbed by wava with a direction of 60° and a speed of 5 m/s as well as the wind with a speed of 5 m/s and direction of 60° at 600 s. The simulation results are shown in Fig. 11, where the triangle represents the initial position of the USV. Similarly,

the LADRC control effect with the parameters of $\omega_o = 0.8$, $\omega_c = 0.008$ is also shown in Fig. 11.

Observing Fig. 11(b), it can be found that under the influence of disturbance, the rudder continuously adjusts its own control amount, which has ensured that the USV can sail along the planned path. Since our guiding law is derived from a straight line, the curve can be traced by approximating a circle into an infinite number of short line segments. However, comparing Figs. 11 and 9, it can be found that although the path tracking control can still be achieved under the influence of wind and waves, the effect of curve tracking is obviously not as good as that of tracking straight lines.

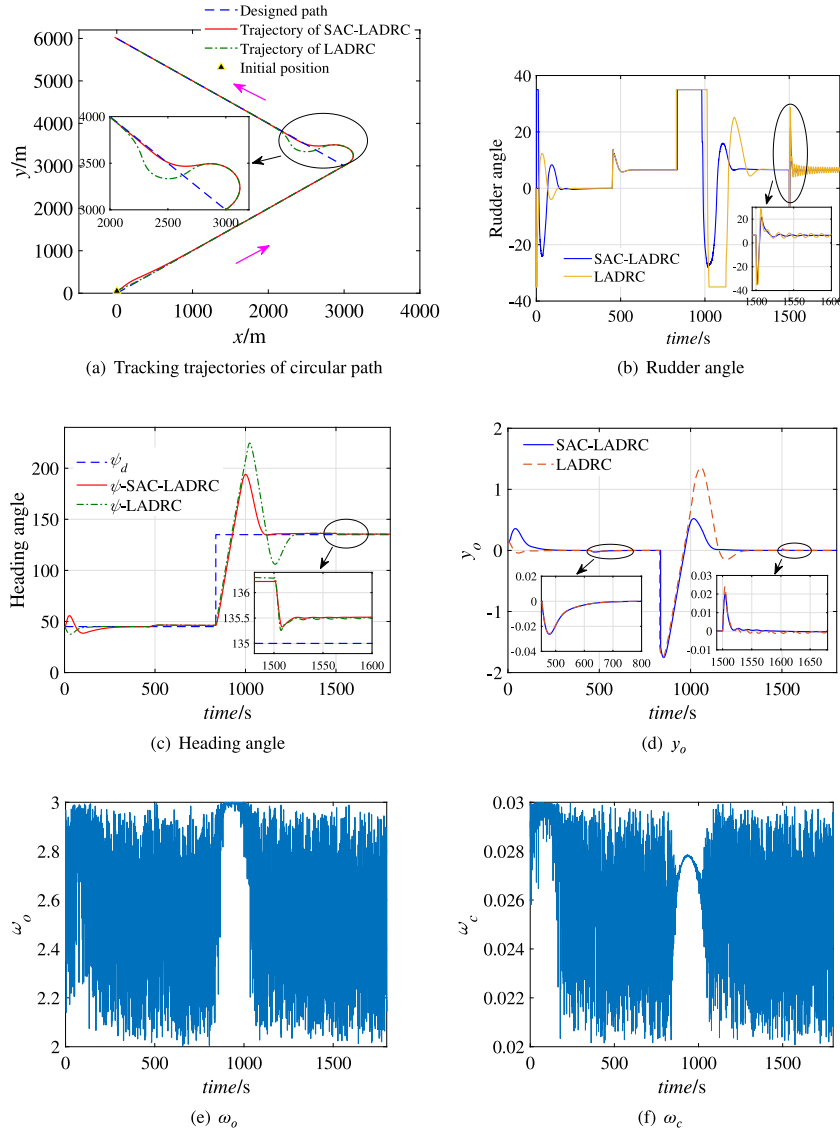


Fig. 9. Tracking results of straight path by SAC-LADRC with wave and wind disturbance.

5.3. Straight and circular path following

The above simulations are all for the same working condition, which shows the effectiveness of the proposed method. In fact, a fundamental reason for choosing SAC as an optimization tool is that it can adaptively select the controller parameters according to the system state so that the control system can have better adaptability. To test the effectiveness of the proposed method under different path conditions, we use a semicircle and a straight line as the planned path. The start point of the semicircle with a radius of 1000 m is (0 m, 1000 m), and the endpoint is (0 m, -1000 m). The semicircle's end point is the line segment's start point, and the endpoint of the line segment is (-1500 m, 1500 m). The simulation results are shown in Fig. 12, where the initial position and heading angle of the USV are (0 m, 1000 m) and 0° , respectively.

To reflect the advantages of the proposed method, the LADRC simulation results with fixed parameters $\omega_o = 0.8$, $\omega_c = 0.008$ are added in Fig. 12. It can be found that in the semicircular part, the difference in tracking effects between them is small, and when the planned path is changed to a straight line, the advantages of SAC-LADRC are reflected. It is conceivable that the parameters in one situation may not be

applicable to another. Therefore, the control effect of LADRC with fixed parameters will be inferior to that of SAC-LADRC.

6. Conclusion

We studied a USV's path following control for underactuated ships based on soft actor-critic (SAC) optimized linear active disturbance rejection control (LADRC). We established a three-DOF USV dynamic model based on MMG, which can describe the plane motion of the USV. A guidance law was proposed, and its stability was proved to use the rudder angle to simultaneously control the USV's position and heading angle. A LADRC was designed to estimate and compensate for disturbances, and its stability was analyzed. To make the controller more robust, we used the SAC algorithm to tune its parameters. SAC has the advantages of a Markov decision process in reinforcement learning, and can use the system state to adaptively select the controller parameters. We selected the state according to the USV's dynamic characteristics and designed an error-based reward function. The effectiveness of the proposed method was verified by the simulation of path tracking on straight lines and circles under wind and water wave disturbances,

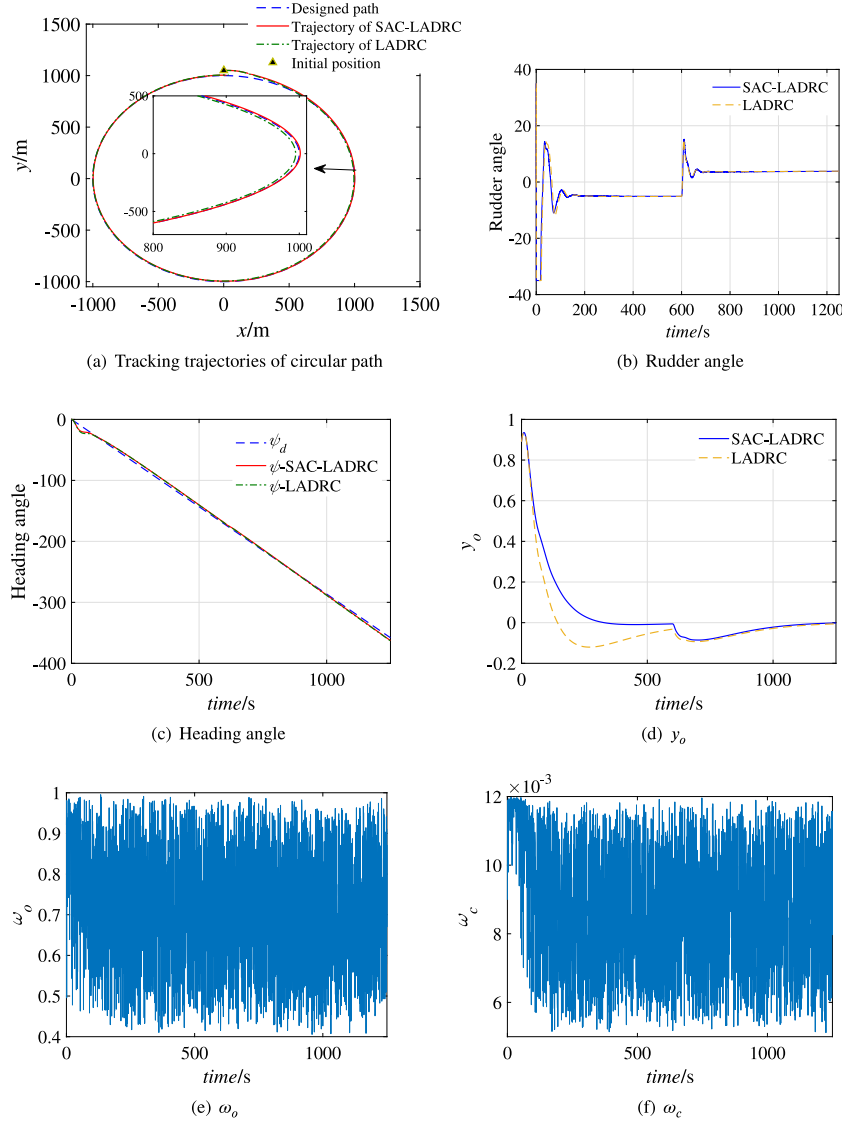


Fig. 10. Tracking results of circular path by SAC-LADRC with wind disturbance.

and the advantages of the proposed method under different working conditions are also reflected by comparison with conventional LADRC.

We studied the plane motion of an underdriven USV, but the heave motion during navigation will also have a certain impact on a USV's path, which we will consider in future work. For the SAC algorithm, although it can indeed help the controller achieve better control performance, we can see that in the steady state, its action changes still very drastically, which is not what we want. Therefore, in the following research, we hope to achieve stable decision-making while retaining the advantages of the algorithm. Finally, applying the proposed method to actual sailing experiments will be our follow-up research.

CRediT authorship contribution statement

Yuemin Zheng: Conceptualization of this study, Methodology, Data curation, Software, Writing – original. **Jin Tao:** Methodology, Writing – review and editing, Supervision, Project administration, Funding acquisition. **Qinglin Sun:** Methodology, Writing – review and editing, Supervision. **Hao Sun:** Investigation, Resources. **Zengqiang Chen:** Conceptualization. **Mingwei Sun:** Conceptualization. **Guangming Xie:** Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No. 61973172, 61973175, 62003175, and 62003177), National Key Research and Development Project (Grant No. 2019YFC1510900), and Key Technologies Research and Development Program of Tianjin (Grant No. 19JCZDJC32800). The project was also funded by the China Postdoctoral Science Foundation (Grant No. 2020M670633) and Academy of Finland under No. 315660.

Appendix

The matrix P has the following elements (p_{35} , p_{44} and p_{45} are given in Boxes I, II and III):

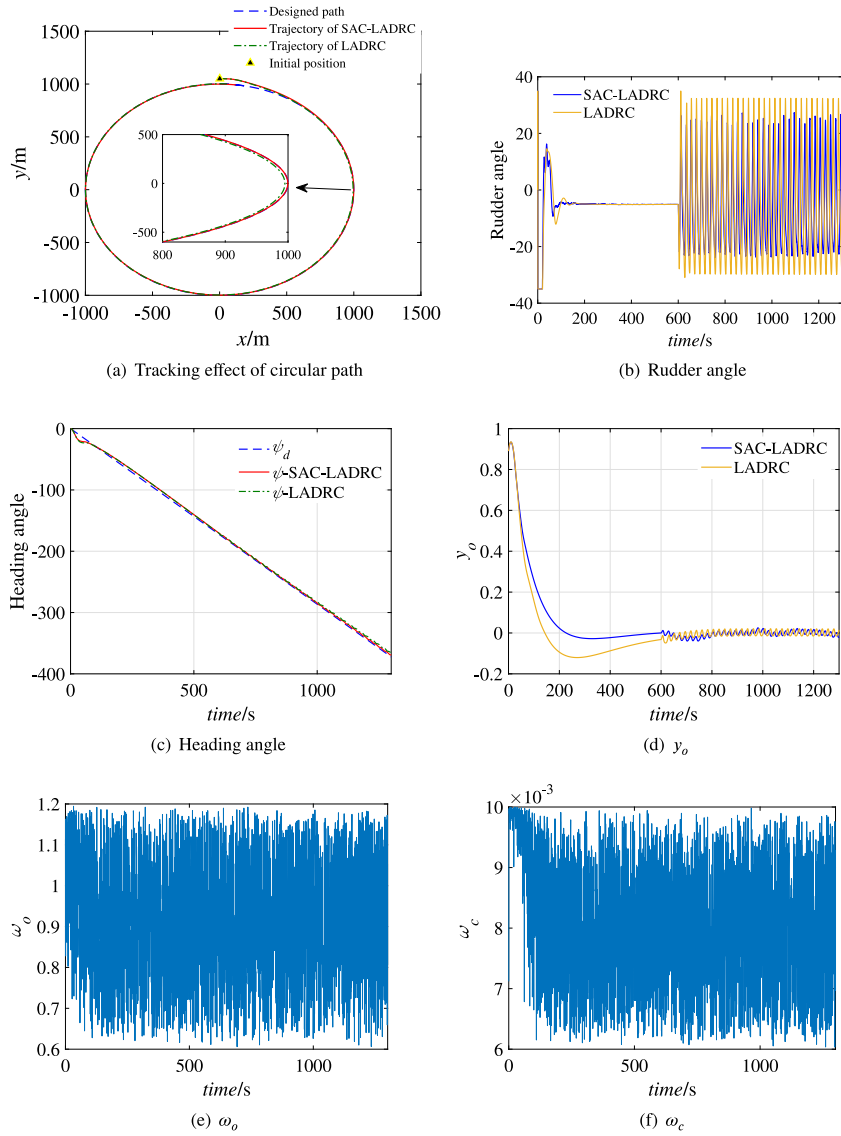


Fig. 11. Tracking results of circular by SAC-LADRC with water wave disturbance.

$$\begin{aligned}
 p_{11} &= \frac{\omega_c^2 + 5}{4\omega_c}, p_{12} = \frac{1}{2\omega_c^2}, \\
 p_{13} &= \frac{\omega_c^6 - 2\omega_c^5\omega_o - 12\omega_c^4\omega_o^2 - 3\omega_c^4 - 2\omega_c^3\omega_o^3 + \omega_c^2\omega_o^4 + 24\omega_c^2\omega_o^2 + 20\omega_c\omega_o^3 + 5\omega_o^4}{4\omega_c(\omega_c + \omega_o)^4}, \\
 p_{14} &= -\frac{-4\omega_c^6 - 7\omega_c^5\omega_o + 6\omega_c^4\omega_o^2 + 10\omega_c^4 + 3\omega_c^3\omega_o^3 + 25\omega_c^3\omega_o + 6\omega_c^2\omega_o^2 - 5\omega_c\omega_o^3 - 2\omega_o^4}{4\omega_c^2(\omega_c + \omega_o)^4}, \\
 p_{15} &= -\frac{-4\omega_c^5 - 10\omega_c^4\omega_o - 3\omega_c^3\omega_o^2 + 8\omega_c^3 + 23\omega_c^2\omega_o + 15\omega_c\omega_o^2 + 3\omega_o^3}{2\omega_c^2(\omega_c + \omega_o)^4}, \\
 p_{22} &= \frac{\omega_c^2 + 1}{4\omega_c^3}, \\
 p_{23} &= \frac{-3\omega_c^5\omega_o - 6\omega_c^4\omega_o^2 - 2\omega_c^4 + 7\omega_c^3\omega_o^3 + \omega_c^3\omega_o + 4\omega_c^2\omega_o^4 + 18\omega_c^2\omega_o^2 + 11\omega_c\omega_o^3 + 2\omega_o^4}{4\omega_c^2(\omega_c + \omega_o)^4},
 \end{aligned}$$

$$\begin{aligned}
 p_{24} &= -\frac{-\omega_c^6 + 2\omega_c^5\omega_o + 12\omega_c^4\omega_o^2 + 7\omega_c^4 + 2\omega_c^3\omega_o^3 + 16\omega_c^3\omega_o - \omega_c^2\omega_o^4 - 4\omega_c\omega_o^3 - \omega_o^4}{4\omega_c^2(\omega_c + \omega_o)^4}, \\
 p_{25} &= -\frac{-4\omega_c^5 - 7\omega_c^4\omega_o + 6\omega_c^3\omega_o^2 + 12\omega_c^3 + 3\omega_c^2\omega_o^3 + 33\omega_c^2\omega_o + 18\omega_c\omega_o^2 + 3\omega_o^3}{4\omega_c^3(\omega_c + \omega_o)(\omega_c^3 + 3\omega_c^2\omega_o + 3\omega_c\omega_o^2 + \omega_o^3)}, \\
 p_{33} &= \frac{\left(4\omega_c^6\omega_o + \omega_c^5\omega_o^4 + 13\omega_c^5\omega_o^2 + 6\omega_c^5 + 4\omega_c^4\omega_o^5 + 60\omega_c^4\omega_o^3 + 16\omega_c^4\omega_o + 6\omega_c^3\omega_o^6 + 184\omega_c^3\omega_o^4 + 82\omega_c^3\omega_o^2 + 4\omega_c^2\omega_o^7 + 100\omega_c^2\omega_o^5 + 216\omega_c^2\omega_o^3 + \omega_c\omega_o^8 + 15\omega_c\omega_o^6 + 116\omega_c\omega_o^4 + 20\omega_o^5\right)}{16\omega_c\omega_o(\omega_c + \omega_o)^4}, \\
 p_{34} &= -\frac{6\omega_c^5\omega_o + 18\omega_c^4\omega_o^2 + 7\omega_c^4 + 6\omega_c^3\omega_o^3 + 16\omega_c^3\omega_o - 4\omega_c\omega_o^3 - \omega_o^4}{2\omega_c^2(\omega_c + \omega_o)^4}, \\
 p_{55} &= \frac{\left(33\omega_c^6\omega_o^4 + 29\omega_c^6\omega_o^2 + 6\omega_c^6 + 99\omega_c^5\omega_o^5 + 54\omega_c^5\omega_o + 99\omega_c^4\omega_o^6 + 165\omega_c^4\omega_o^4 + 252\omega_c^4\omega_o^2 + 33\omega_c^3\omega_o^7 + 155\omega_c^3\omega_o^5 + 576\omega_c^3\omega_o^3 + 36\omega_c^3\omega_o + 612\omega_c^2\omega_o^4 + 252\omega_c^2\omega_o^2 + 36\omega_o^6\right)}{16\omega_c^3\omega_o^2(\omega_c^3\omega_o^3 + 3\omega_c^2\omega_o^4 + 3\omega_c\omega_o^5 + \omega_o^6)},
 \end{aligned}$$

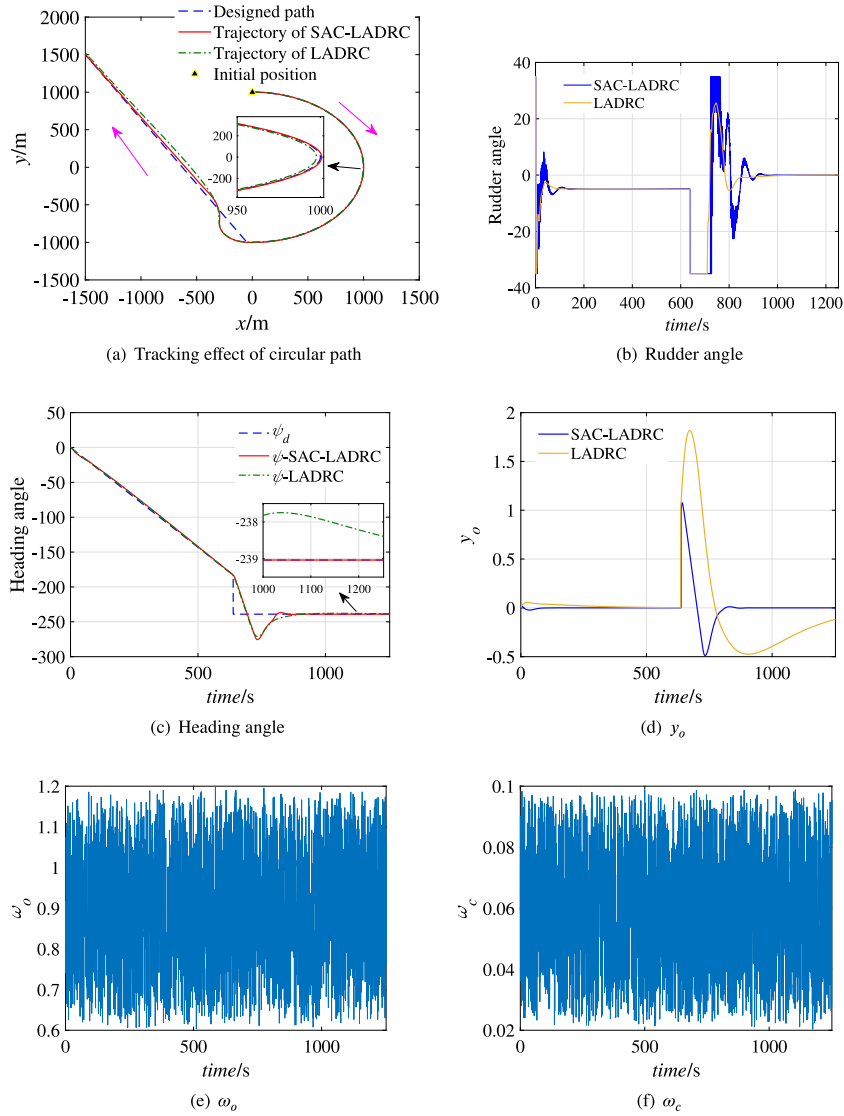


Fig. 12. Tracking results of circular and straight path by SAC-LADRC.

$$p_{35} = - \frac{(3\omega_c^6\omega_o^4 + 15\omega_c^6\omega_o^2 + 2\omega_c^6 + 12\omega_c^5\omega_o^5 + 64\omega_c^5\omega_o^3 + 20\omega_c^5\omega_o + 18\omega_c^4\omega_o^6 + 136\omega_c^4\omega_o^4 + 102\omega_c^4\omega_o^2 + 12\omega_c^3\omega_o^7 + 156\omega_c^3\omega_o^5 + 276\omega_c^3\omega_o^3 + 3\omega_c^2\omega_o^8 + 45\omega_c^2\omega_o^6 + 348\omega_c^2\omega_o^4 + 156\omega_c\omega_o^5 + 24\omega_o^6)}{16\omega_c^2\omega_o^3(\omega_c + \omega_o)^4}$$

Box I.

$$p_{44} = \frac{(3\omega_c^7\omega_o^4 + 15\omega_c^7\omega_o^2 + 2\omega_c^7 + 12\omega_c^6\omega_o^5 + 100\omega_c^6\omega_o^3 + 20\omega_c^6\omega_o + 18\omega_c^5\omega_o^6 + 184\omega_c^5\omega_o^4 + 102\omega_c^5\omega_o^2 + 12\omega_c^4\omega_o^7 + 60\omega_c^4\omega_o^5 + 152\omega_c^4\omega_o^3 + 3\omega_c^3\omega_o^8 + 13\omega_c^3\omega_o^6 + 20\omega_c^3\omega_o^4 + 4\omega_c^2\omega_o^7 + 12\omega_c^2\omega_o^5 + 16\omega_c\omega_o^6 + 4\omega_o^7)}{16\omega_c^3\omega_o^3(\omega_c + \omega_o)^3}$$

Box II.

$$p_{45} = -\frac{2\omega_c^7 + 8\omega_c^6\omega_o + 12\omega_c^5\omega_o^2 - 4\omega_c^5 + 8\omega_c^4\omega_o^3 - 7\omega_c^4\omega_o + 2\omega_c^3\omega_o^4 + 6\omega_c^3\omega_o^2 + 12\omega_c^3 + 3\omega_c^2\omega_o^3 + 33\omega_c^2\omega_o + 18\omega_c\omega_o^2 + 3\omega_o^3}{4\omega_c^3(\omega_c + \omega_o)^4}$$

Box III.

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